REAR-IR1: rear docking IR board, 1 3.3 V, 2 IR_L (PC8, TIM3_CH3), 3 IR_R (PC9, TIM3_CH4), 4 GND.
HALL-BIN1 / HALL-TOWER1: Hall boards, SH 3-pin (ZX-SH1.0-3PLT, so a Hall cable cannot go into a 4-pin socket): 1 3.3 V, 2 OUT (PE4 / PE5, low = magnet near), 3 GND. All mounting pads to GND.

CLIFF
File: CLIFF.kicad_sch

BUMPER
File: bumper_sensors.kicad_sch

MAIN BRUSH MOTOR
File: MAIN-BRUSH-MOTOR .kicad_sch

WHEEL MOTORs
File: WHEEL-MOTORs .kicad_sch

MAIN FAN
File: MAIN-FAN .kicad_sch

SIDE BRUSH MOTORs
File: SIDE-BRUSH-MOTORs .kicad_sch

MCU-STM32
File: STM32G473.kicad_sch

CARPET-SENSOR
File: Carpet-sensor.kicad_sch

WD
File: WATCHDOG.kicad_sch

LiDAR
File: LiDAR.kicad_sch

FRONT SENSORS
File: Front-Sensors.kicad_sch

IMU
File: IMU-ICM-4267-P .kicad_sch

BUTTON-LEDs
File: BUTTON-LEDs .kicad_sch

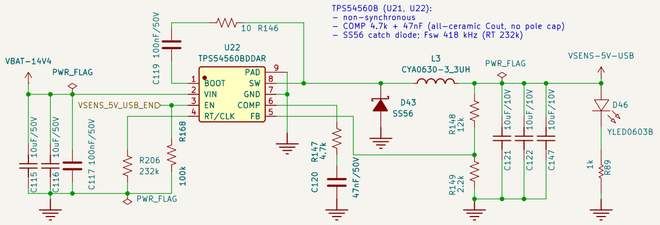
POWER CONTROL
File: SYSTEM-POWER.kicad_sch

BATTERY CHARGER
File: Battery-Charger.kicad_sch

Sheet: / File: Main.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 1/16

5V USB, SENSORS, LIDAR POWER RAIL
– set 5V_EN HIGH to enable

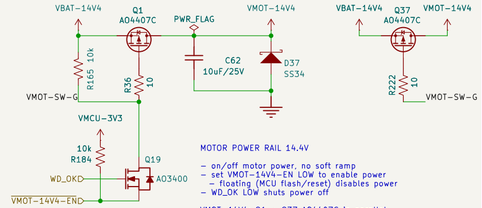
TPS54560B (U21, U22):
– non-synchronous
– COMP 4.7k + 47nF (all-ceramic Cout, no pole cap)
– SS56 catch diode; Fsw 418 kHz (RT 232k)



MOTOR POWER RAIL 14.4V

– on/off motor power, no soft ramp
– set VMOT-14V4-EN LOW to enable power
  – floating (MCU flash/reset) disables power
– WD_OK LOW shuts power off

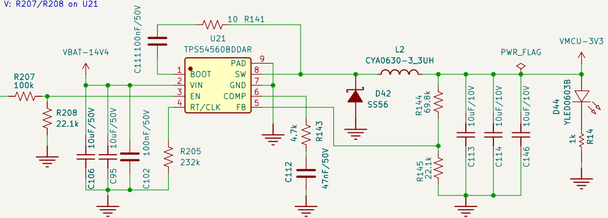
VMOT-14V4: Q1 + Q37 AO4407C in parallel
– ~7.5 A normal (with the MSD-G fan) is ~0.25 W each
– short stall peaks (~13 A) fine with copper under the drain
– Vgs +/-25 V, no gate divider needed



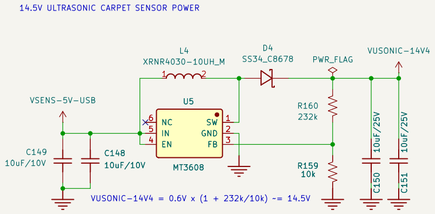
3.3V MCU POWER RAIL

– POWER_EN is 14.4V
– set POWER_EN HIGH to enable

TPS54560B (U21, U22):
– non-synchronous
– COMP 4.7k + 47nF (all-ceramic Cout, no pole cap)
– SS56 catch diode: fsw 418 kHz (RT 232k)
– EN abs max 8.4 V: R207/R208 on U21



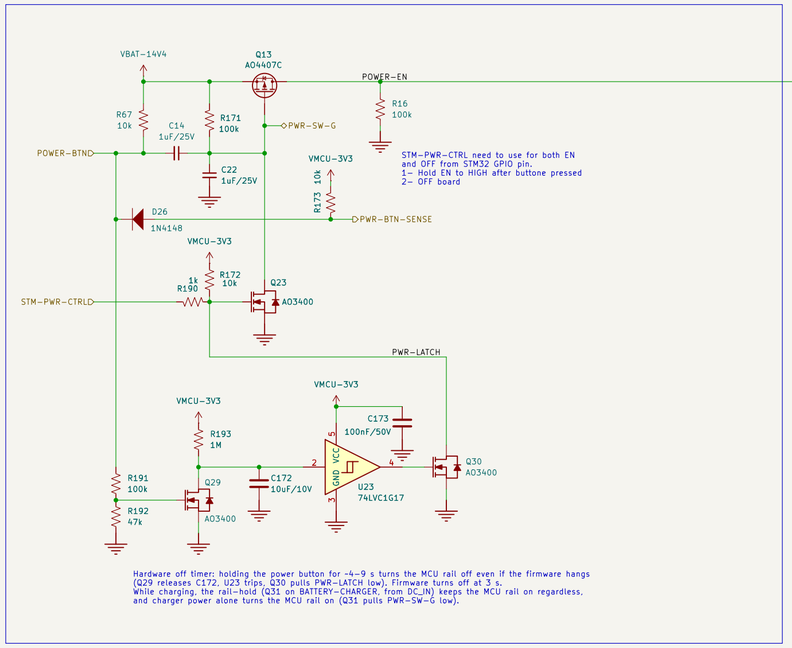
14.5V ULTRASONIC CARPET SENSOR POWER



VUSONIC-14V4 = 0.6V x (1 + 232k/10k) ~= 14.5V

STM-PWR-CTRL need to use for both EN and OFF from STM32 GPIO pin.
1– Hold EN to HIGH after buttone pressed
2– OFF board

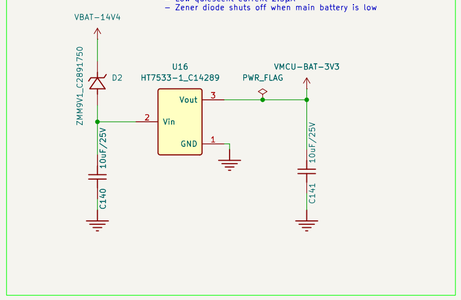
Hardware off timer: holding the power button for ~4–9 s turns the MCU rail off even if the firmware hangs (Q29 releases C172, U23 trips, Q30 pulls PWR-LATCH low), Firmware turns off at 3 s.
While charging, the rail-hold (Q31 on BATTERY-CHARGER, from DC_IN) keeps the MCU rail on regardless, and charger power alone turns the MCU rail on (Q31 pulls PWR-SW-G low).



3.3V SENSORS POWER RAIL

– C176 keeps Q16 off when power switches ON
– C64 soft-ramps power
– set VSENS_3V3_EN LOW to enable sensors power

MCU VBAT
– Low quiescent current 2.5µA
– Zener diode shuts off when main battery is low



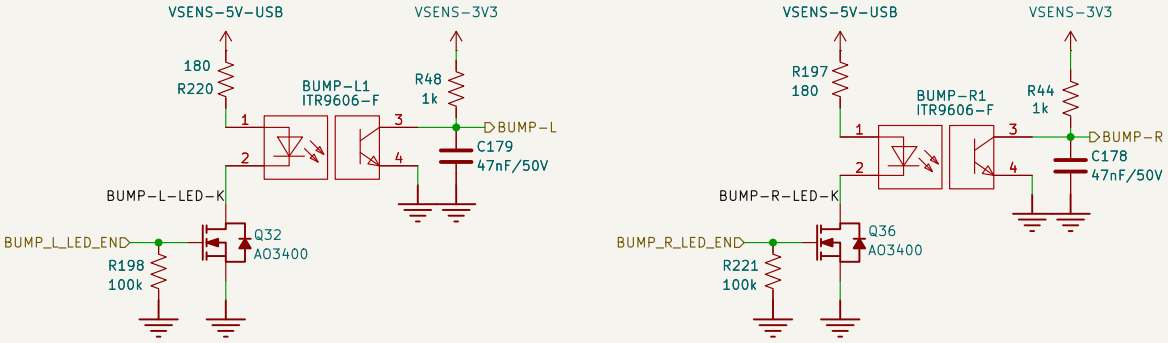
PCB mounting standoffs (Wurth WA-SMSI, M3), moved here from the removed M.2 sheet.



Sheet: /POWER/
File: SYSTEM-POWER.kicad_sch

Title:

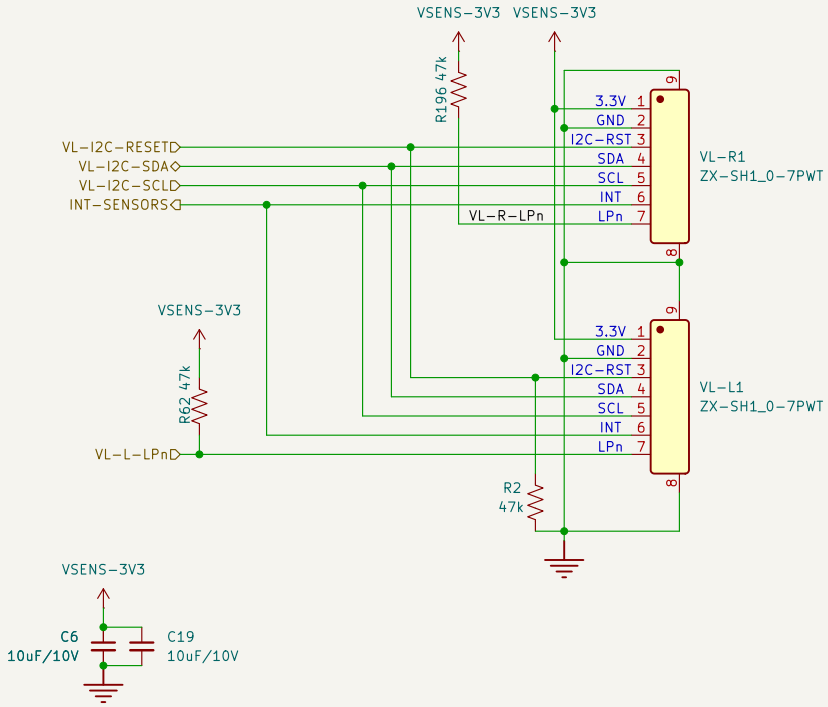
Size: A2 | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 2/16



- slotted photo-interrupters
- BUMP-X reads low when light passes the slot
- drive BUMP_L/R_LED_EN PWM <<15 usec period to enable
- tune R44, R48, LED PWM duty to let ADC sense full open/close range

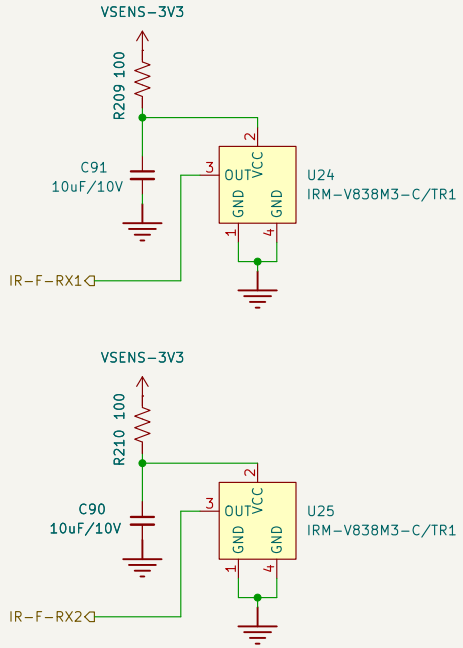
Sheet: /Bumper Sensors/	
File: bumper_sensors.kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 3/16

2x VL53L7 ToF BOARDS



2 IR RECEIVERS

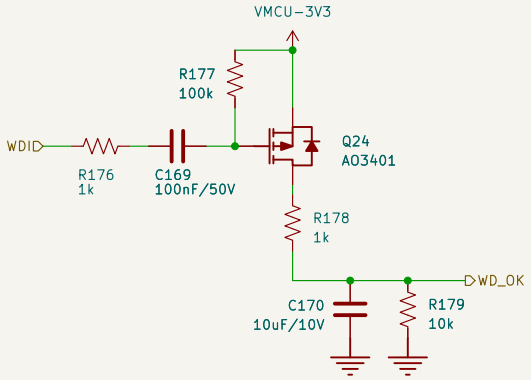
U24 left, U25 right (robot view). Side-looking: put both at the board front edge, lenses facing forward, ~10 mm apart, with a baffle between them in the bumper window. – 90 deg FoV



Sheet: /FRONT SENSORS/
File: Front-Sensors.kicad_sch

Title:

Size: USLetter	Date:	**Rev:**
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RC CHARGE-PUMP WATCHDOG

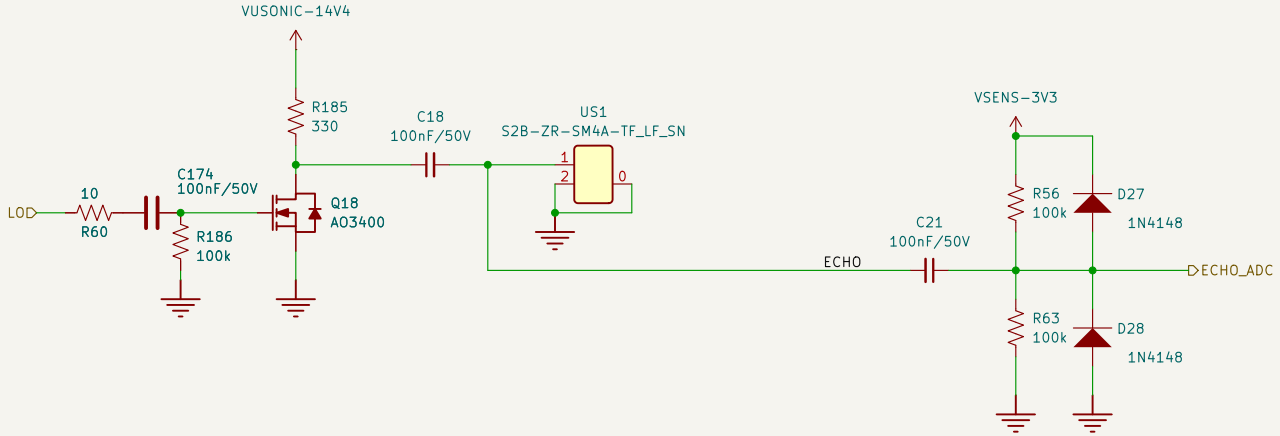
- WD_OK active HIGH
- WDI HIGH->LOW transition pets the dog

Firmware toggles WDI in software (50-1000 Hz, ~50% duty) only while the control loop is healthy.
Never drive WDI from a timer/PWM: a timer keeps running after the CPU hangs.
Each falling edge on WDI pulls the Q24 gate low through C169 and tops up C170 via R178.
WD_OK ~2.7 V while toggling. WDI stuck high, low or floating: <1 V in ~100 ms, <0.4 V in ~190 ms
(10 uF nominal; DC-bias derating shortens this). WD_OK starts at 0 V at power-up (fail-safe).
WD_OK drives N-FET gates only (14.4 V rail switch, LiDAR incl. laser): no resistive loads.
WDI pin: high-Z at reset, no pull-down.

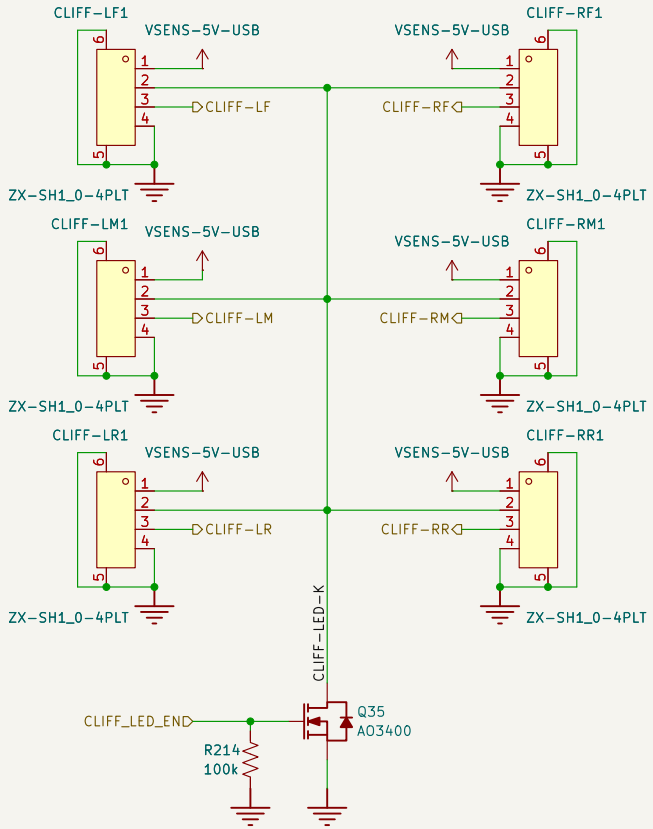
Sheet: /WATCHDOG/	
File: WATCHDOG.kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 10.0.4	Id: 5/16

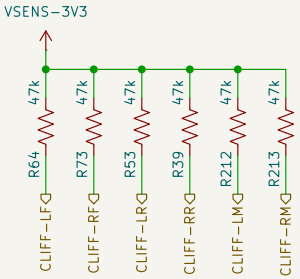
Carpet sensor driver: one low-side AO3400 + 330R pull-up from VUSONIC-14V4 (14.4 V). The P-FET half-bridge was removed: its 100k gate pull-up could not turn it off at 290 kHz (shoot-through). R186 holds LO low in reset/flash.
Drive LO in short bursts only: LO stuck high puts 0.63 W into R185.

C174 AC-couples LO to the Q18 gate: if LO sticks high, the gate decays through R186 (~10 ms)
and Q18 turns off, so R185 is never stressed by firmware holding LO high.



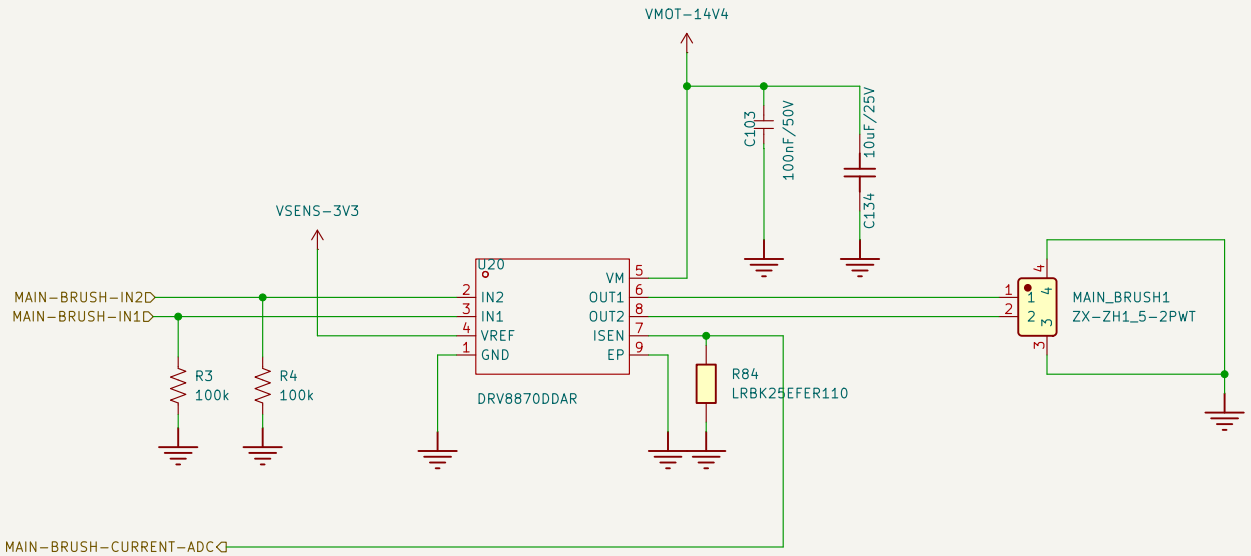
Sheet: /CARPET-SENSOR/		
File: Carpet-sensor.kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 6/16



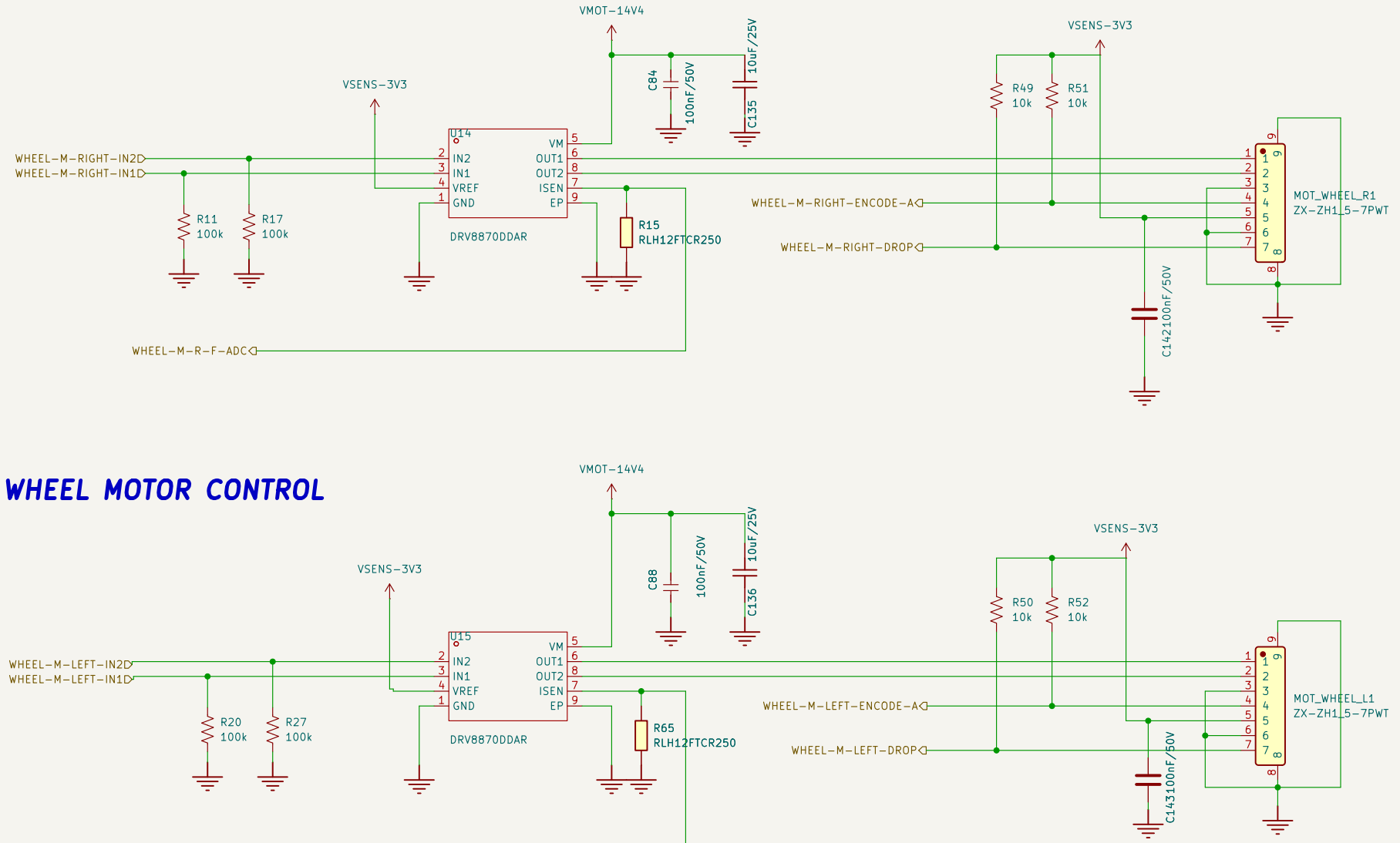


- kicad/cliff-sensor mating boards
- SH 4-pin vertical; use same (pin 1 to pin 1) cable
- set CLIFF_LED_EN LOW to disable
- modulate CLIFF_LED_EN, read each channel LED-on minus LED-off to cancel ambient light

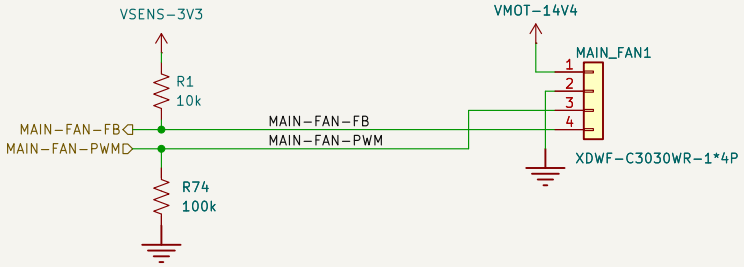
Sheet: /CLIFF/	
File: CLIFF.kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 19/16



Sheet: /MAIN-BRUSH MOTOR/	
File: MAIN-BRUSH-MOTOR .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 22/16



Sheet: /WHEEL MOTOR/		
File: WHEEL-MOTORs .kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 23/16

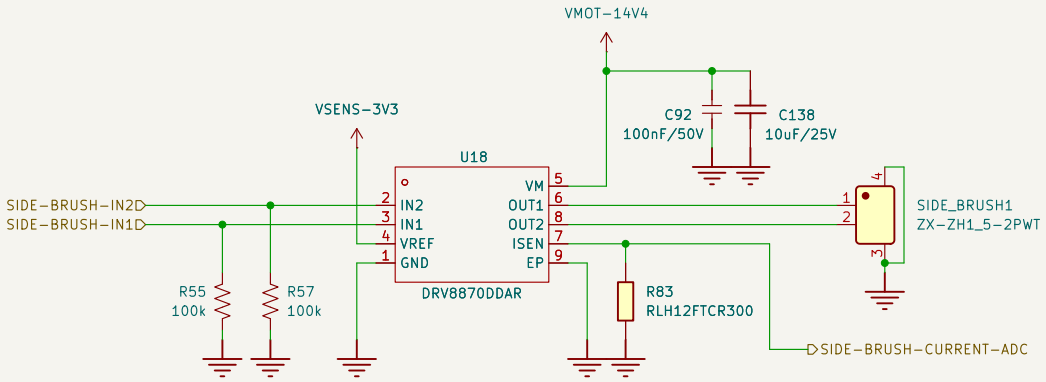


- FB feedback signal is open collector
- set PWM LOW to turn fan OFF
- MAIN_FAN1 matches Dreame L60 PE Ultra PCB

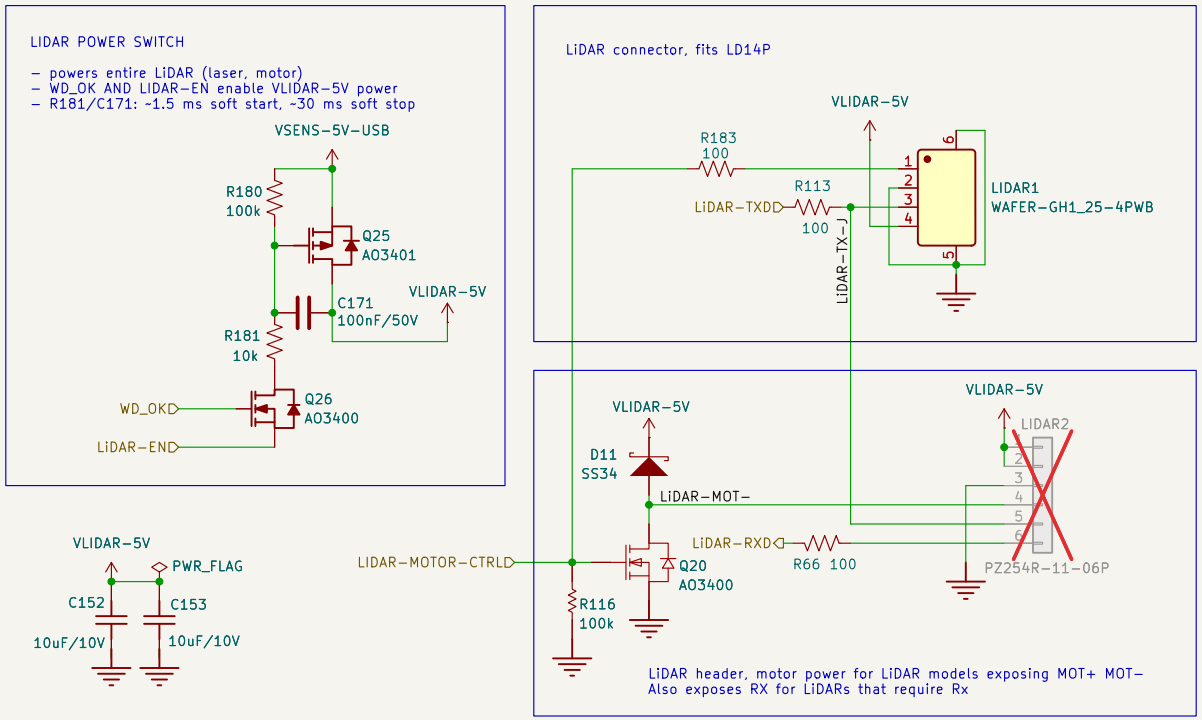


Sheet: /MAIN-FAN/	
File: MAIN-FAN .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 24/16

SIDE-BRUSH MOTOR CONTROL

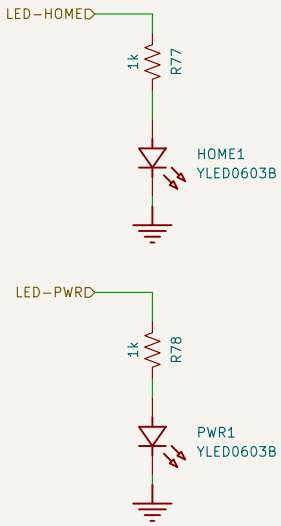


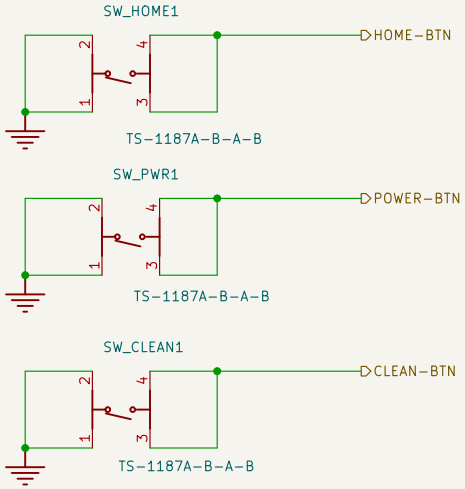
Sheet: /SIDE-BRUSH/		
File: SIDE-BRUSH-MOTORs .kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 25/16



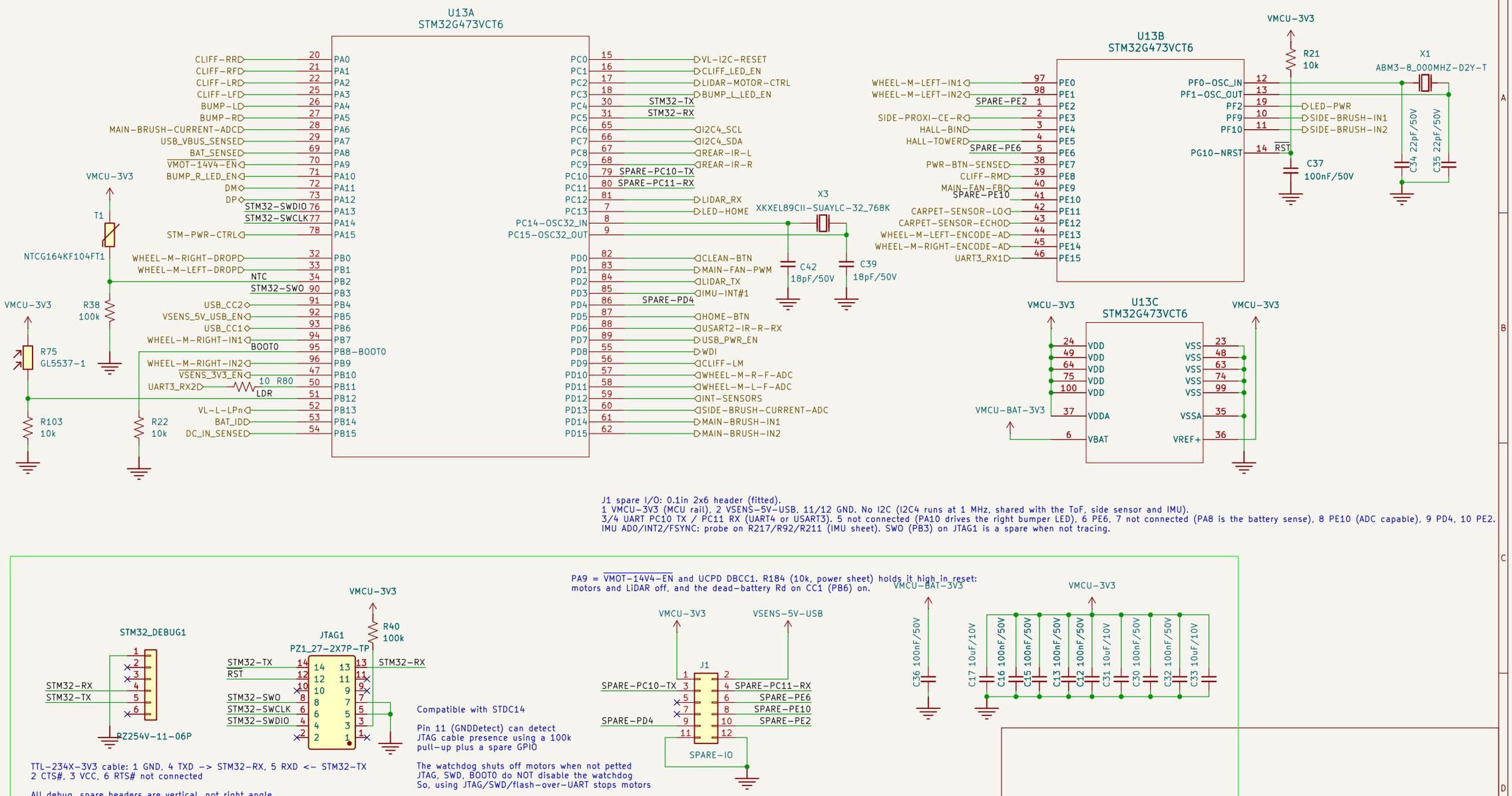
Sheet: /LiDAR/	
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Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 26/16

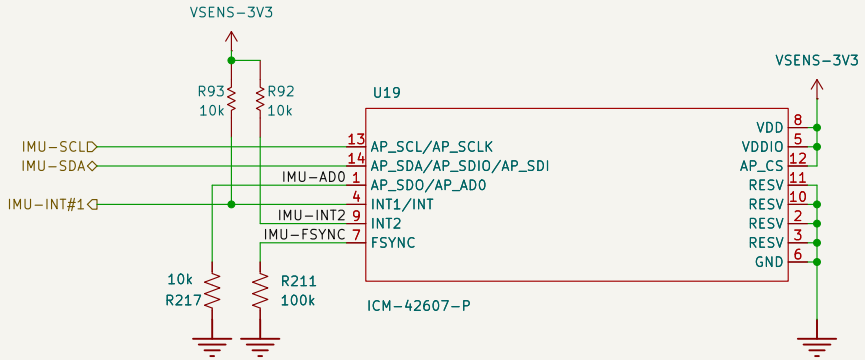
LEDs AND BUTTONs



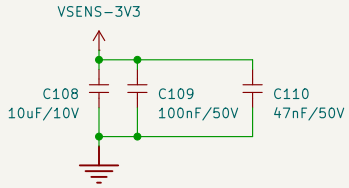


Sheet: /BUTTON-LED/ File: BUTTON-LEDs .kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 28/16


U13A
STM32G473VCT6
U13B
STM32G473VCT6
U13C
STM32G473VCT6
VMCU-3V3
VMCU-BAT-3V3
VSENS-5V-USB
T1
NTCG164KF104FT1
R38
100k
R75
GL5537-1
R103
10k
R22
10k
R40
100k
R21
10k
X1
ABM3-8_000MHZ-D2Y-T
X3
XKXEL89CII-SUAYLC-32_768K
C42
18pF/50V
C39
18pF/50V
C37
100nF/50V
C34 22pF/50V
C35 22pF/50V
C36 100nF/50V
C17 10uF/10V
C16 100nF/50V
C15 100nF/50V
C13 100nF/50V
C12 100nF/50V
C31 10uF/10V
C30 100nF/50V
C32 100nF/50V
C33 10uF/10V
CLIFF-RR
CLIFF-RF
CLIFF-LR
CLIFF-LF
BUMP-LD
BUMP-RD
MAIN-BRUSH-CURRENT-ADC
USB_VBUS_SENSED
BAT_SENSED
VMOT-14V4-EN
BUMP_R_LED_EN
DM
DP
STM32-SWDIO
STM32-SWCLK
STM-PWR-CTRL
WHEEL-M-RIGHT-DROP
WHEEL-M-LEFT-DROP
NTC
STM32-SWO
USB_CC2
VSENS_5V_USB_EN
USB_CC1
WHEEL-M-RIGHT-IN1
BOOT0
WHEEL-M-RIGHT-IN2
VSENS_3V3_EN
UART3_RX2
10 R80
LDR
VL-L-LPn
BAT_ID
DC_IN_SENSED
VL-I2C-RESET
CLIFF_LED_EN
LIDAR-MOTOR-CTRL
BUMP_L_LED_EN
STM32-TX
STM32-RX
I2C4_SCL
I2C4_SDA
REAR-IR-L
REAR-IR-R
SPARE-PC10-TX
SPARE-PC11-RX
LIDAR_RX
LED-HOME
PC14-OSC32_IN
PC15-OSC32_OUT
CLEAN-BTN
MAIN-FAN-PWM
LIDAR_TX
IMU-INT#1
SPARE-PD4
HOME-BTN
USART2-IR-R-RX
USB_PWR_EN
WDI
CLIFF-LM
WHEEL-M-R-F-ADC
WHEEL-M-L-F-ADC
INT-SENSORS
SIDE-BRUSH-CURRENT-ADC
MAIN-BRUSH-IN1
MAIN-BRUSH-IN2
WHEEL-M-LEFT-IN1
WHEEL-M-LEFT-IN2
SPARE-PE2
SIDE-PROXI-CE-R
HALL-BIND
HALL-TOWERD
SPARE-PE6
PWR-BTN-SENSED
CLIFF-RMD
MAIN-FAN-FBD
SPARE-PE10
CARPET-SENSOR-LO
CARPET-SENSOR-ECHO
WHEEL-M-LEFT-ENCODE-AD
WHEEL-M-RIGHT-ENCODE-AD
UART3_RX1
PF0-OSC_IN
PF1-OSC_OUT
PF2
PF9
PF10
PG10-NRST
LED-PWR
SIDE-BRUSH-IN1
SIDE-BRUSH-IN2
RST
VDD
VSS
VDDA
VSSA
VBAT
VREF+
J1 spare I/O: 0.1in 2x6 header (fitted).
1 VMCU-3V3 (MCU rail), 2 VSENS-5V-USB, 11/12 GND. No I2C (I2C4 runs at 1 MHz, shared with the ToF, side sensor and IMU).
3/4 UART PC10 TX / PC11 RX (UART4 or USART3). 5 not connected (PA10 drives the right bumper LED), 6 PE6, 7 not connected (PA8 is the battery sense), 8 PE10 (ADC capable), 9 PD4, 10 PE2.
IMU AD0/INT2/FSYNC: probe on R217/R92/R211 (IMU sheet). SWO (PB3) on JTAG1 is a spare when not tracing.
PA9 = VMOT-14V4-EN and UCPD DBCC1. R184 (10k, power sheet) holds it high in reset:
motors and LiDAR off, and the dead-battery Rd on CC1 (PB6) on.
STM32_DEBUG1
PZ254V-11-06P
JTAG1
PZ1_27-2X7P-TP
J1
SPARE-IO
SPARE-PC10-TX
SPARE-PD4
SPARE-PC11-RX
SPARE-PE6
SPARE-PE10
SPARE-PE2
Compatible with STDC14
Pin 11 (GNDDetect) can detect
JTAG cable presence using a 100k
pull-up plus a spare GPIO
The watchdog shuts off motors when not petted
JTAG, SWD, BOOT0 do NOT disable the watchdog
So, using JTAG/SWD/flash-over-UART stops motors
TTL-234X-3V3 cable: 1 GND, 4 TXD -> STM32-RX, 5 RXD <- STM32-TX
2 CTS#, 3 VCC, 6 RTS# not connected
All debug, spare headers are vertical, not right angle
Sheet: /MCU-STM32/
File: STM32G473.kicad_sch
Title:
Size: USLegal
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 30/16

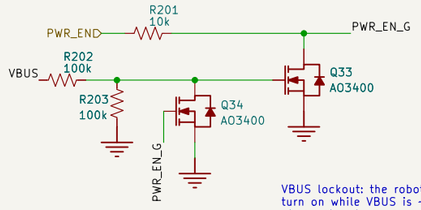


IMU on I2C4 (2.2k pull-ups R5/R6 on the top sheet). AP_AD0 pulled low by R217 (I2C address 0x68).
The IMU is leadless: probe AD0 on R217, INT2 on its pull-up R92 and FSYNC on R211. Run a wire to a spare pin if INT2 or FSYNC is needed.



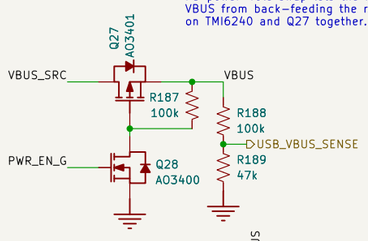
Sheet: /IMU/	
File: IMU-ICM-4267-P .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 32/16

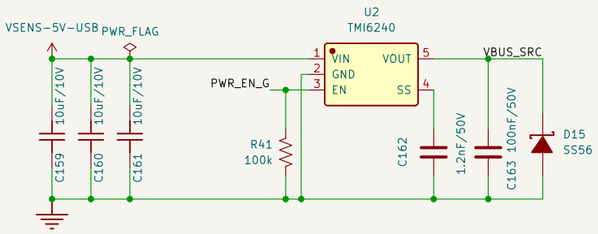
USB-C

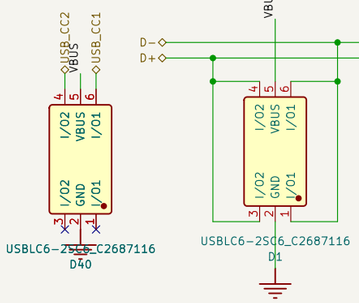


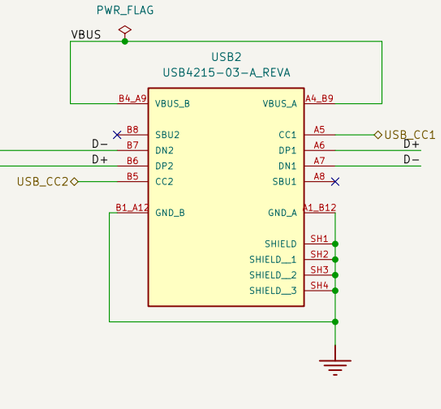
VBUS lockout: the robot 5 V (TMI6240 + Q27) can only turn on while VBUS is ~0 V. Q33 holds PWR_EN_G low if a phone already sources VBUS; once on, Q34 latches it.

USB-C (phone): phone is the USB host. CC1/CC2 go to the STM32 UCPD (PB6/PB4): Rd by default, PD power-role swap lets the robot charge the phone while it stays host. Q27 blocks the phone's VBUS from back-feeding the robot 5 V (the TMI6240 body diode conducts VOUT->VIN); PWR_EN turns on TMI6240 and Q27 together. TMI6240 now runs from VSENS-5V-USB (U25 buck removed).

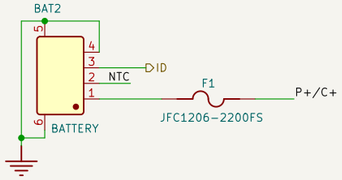




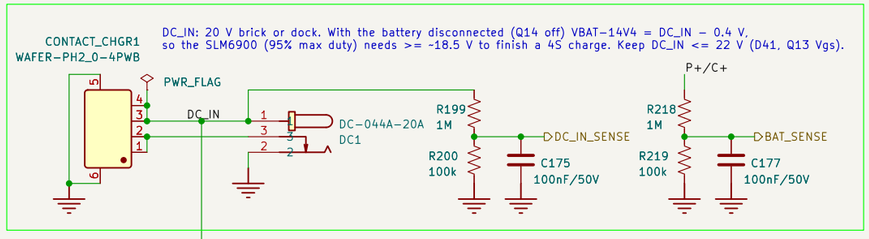




BATTERY CONNECTOR

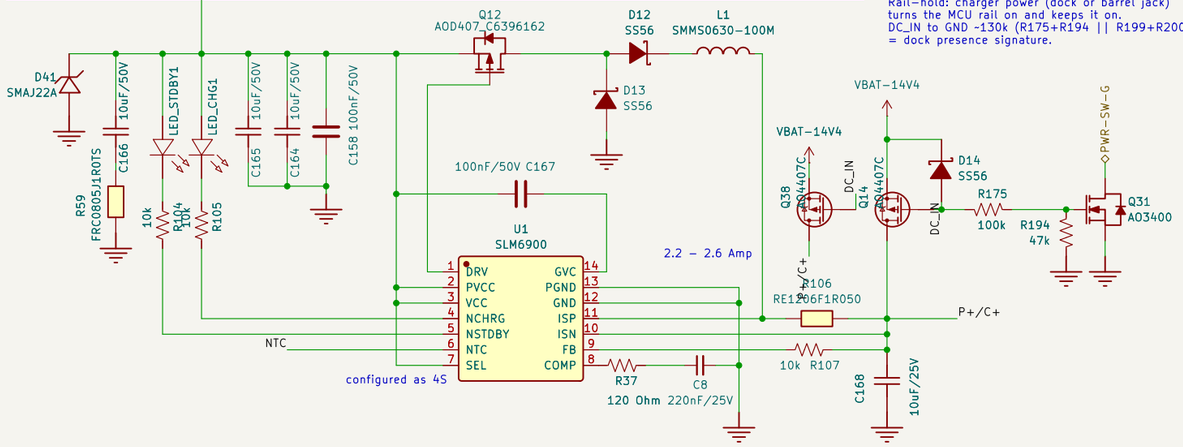


BATTERY CHARGER



DC_IN: 20 V brick or dock. With the battery disconnected (Q14 off) VBAT-14V4 = DC_IN - 0.4 V, so the SLM6900 (95% max duty) needs >= ~18.5 V to finish a 4S charge. Keep DC_IN <= 22 V (D41, Q13 Vgs).

Rail-hold: charger power (dock or barrel jack) turns the MCU rail on and keeps it on. DC_IN to GND ~130k (R175+R194 || R199+R200) = dock presence signature.



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